

Friction Force Measurement at Brake Discs

2011-01-1576

Published
05/17/2011

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doi:10.4271/2011-01-1576

ABSTRACT

Experimental researches on brake squeal have been performed since many years in order to get an insight into friction-excited vibrations and squeal triggering mechanisms. There are many different possibilities to analyse brake squeal. The different operating deflection shapes can be detected using e.g. laser vibrometer systems or acceleration sensors. Piezoelectric load cells can be used for the measurement of the normal contact force of the brake pad.

The presented test setup measures not only the mean value of the friction force between brake pad and disc at a certain brake pressure, but also the superposed vibration of this force, which only occurs during a squeal event. Therefore the guide pins of the brake caliper are replaced by modified ones. The brake pads are held in position by these pins and the resulting force of the brake torque, hence the friction force, acts on these pins. The shape of the pins is optimized for measuring these forces. For the measurement strain gauges are mounted on the pin. The dominant frequency of this measured vibration correlates with the microphone used in parallel.

Simultaneously to the test runs a simple analytical model has been developed, which represents the tested brake system. Further research on the test rig regarding the squeal triggering mechanism and the contact between disc and pad is expected to provide new insight into the elusive problem of brake squeal. This will be used to adapt the simple analytical model. The aim is to develop an experimentally verified contact algorithm, which can be used in simplified multi-body simulations or detailed finite element analyses. Thus, it will be possible to predict the critical squeal frequencies of a new brake system and to estimate the effect of potential countermeasures.

INTRODUCTION

A brake that squeals does not indicate a poor brake performance, but this annoying noise causes very high warranty costs. For example Akay [1] declared that in North America the estimated warranty costs for brake NVH issues reach one billion dollars every year. During the engineering of a vehicle the brake squeal issue is detected after changes in the production process were done or just before *Start of Production* in the prototype phase. This is mainly because at this point road tests with the entire vehicle are possible. In order to minimize the occurrence of brake squeal, expensive and time consuming trial and error methods are used most of the time. Thus other methods should be developed to reduce the likelihood of brake squeal earlier.

Therefore, a well-founded contact algorithm for the friction effects between brake pad and disc is absolutely necessary. Thereby it does not matter whether an analytical model is simulated or calculations using the Finite Elements Method (FEM) are performed. But such a new developed contact algorithm has to be verified by doing measurements on an existing brake system that can be simulated afterwards.

The majority of the used measurement methods, which can be found in the literature, measure the displacement, velocity or the acceleration. Thus it is possible to detect operating deflection shapes of brake system components, e.g. the brake disc or the brake caliper. Therefore, a laser vibrometer can be used that measures the velocity of predefined points. See [2] for a laser metrology overview and [3] for a brake squeal measurement system application example. Other possibilities are inductive or capacitive sensors which measure the displacement of the brake disc. For example Ouyang et al. [4] used twelve capacitive sensors which are arranged tangentially at a certain diameter close to the brake disc. Alternatively Sardá [5] described a measuring system with

five capacitive sensors which are arranged in radial direction for measuring the disc displacement. Marschner [6] introduced a measurement system with a triaxial acceleration sensor, which is mounted inside the disc vents. Another approach is the measurement of the acting force between brake disc and pad in normal direction. Therefore, the piezoelectric sensors are mounted in the brake pads [7] or on the brake cylinders [8]. An advantage of these piezoelectric sensors is that they also can be used as actuators. Hence it is possible to suppress the vibration with those actuators. Strain gauges can be used for friction force measurement too [9].

For the selection of the right measuring system the operating conditions have to be considered. During braking high temperatures occur in the contact interface between brake disc and pads. This heat causes a change of temperature at the components in the brake system, especially at those near to the brake disc or pad. These resulting temperatures can be calculated with FEM models, e.g. [10]. Metal strain gauges have a temperature drift that can be compensated using a full bridge circuit. Piezoelectric load cells have a high temperature drift and because of that it is necessary to use elaborate measures when measuring the force vibration at the hot brake pad.

This paper is about using strain gauges for force measurement. The strain gauges are located at the guide pins, which support the acting brake force and brake torque, respectively. The guide pins are redesigned, so that it is possible to measure the force in principle and the superposed vibration, which only occurs during squeal events. Thus this vibration is generated by exciting effects in the contact area between brake disc and pad. These special guide pins are the main part of the developed measurement system. With the measurement results, a phenomenological contact algorithm will be developed and verified.

METHODS

The available brake and suspension test rig offers optimal setup for repeatable analyses. There are three possible test setups: First it is possible to test only the brake system (disc, caliper and pad) on a flywheel test rig. The second variant is testing the corner including parts of the axle and the wheel. And the third possibility is testing a complete vehicle axle which is mounted on a special rack. This rack has the same connection points to the axle as the car body has. The whole test rig is located in a climatic chamber. Hence it is possible to simulate various environmental conditions like hot or cold temperature or high and low humidity.

With the different test setups it is among others possible to analyse the friction behaviour and performance of different pad materials or to evaluate the influence of environmental conditions, e.g. to analyse “morning squeal” which usually occurs in cold and humid conditions. Furthermore the test rig

can operate automatically, thus brake performance tests (SAE J2522) and brake noise tests (SAE J2521) can be run autonomous. Fig. 1 upper shows exemplary a test setup of the brake test rig. This setup was used for a performance test. Because of that thermocouples are mounted in the pads and in the disc. The signals from the rotating disc are transferred using telemetry. The drive shaft from the electric motor is located on the right side. Fig. 1 lower shows exemplary the suspension test rig. With this setup it is possible to test a suspension under almost road conditions. Hence the acting forces between tire and road can be simulated on the test rig. Thus for example endurance tests can be performed.

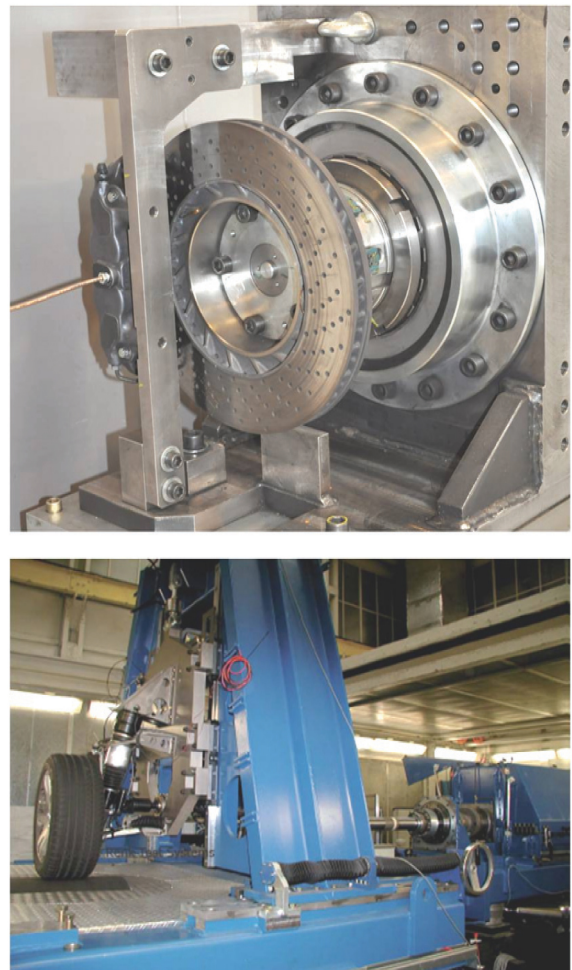


Figure 1. Brake and suspension test rig.

The presented analyses are performed on a special rack, thus the test setup includes the complete axle. The axle consists of prototype parts. Hence, it is not possible to show pictures of the special rack on the test rig. Road tests with prototype vehicles pointed out, that there is a squeal issue at about 2 kHz. The first experiments were done at the brake test rig, hence without the axle. But the main result of these first experiments, which only includes brake disc, pad and caliper, was, that it is not possible to generate squeal. This is because

the boundary conditions are too stiff at the brake test rig. Because of that the whole vehicle axle has been build up on the special rack.

The chosen approach was to measure the friction force with a high resolution and a high sampling rate. For that it is necessary to measure this force as near as possible at the excitation area. The brake torque is generated by the friction force between brake pad and disc. This force is influenced by the pad shape, the contact pressure distribution, the contact pressure, the pad material, the disc material and by the operating and environmental conditions. [Fig. 2](#) shows a simplified illustration of the brake disc and pad containing geometrical information. The blue marked area A shows the contact area. Generated by the disc rotation ω the friction force $\tau(r, \varphi)$ loads the brake pad. The forces F_1 and F_2 are the resulting forces of the pads guide pins and F_3 of the locking pin, which hold the pad in position. The normal force between brake pad and disc, which is necessary for the friction force, is generated by hydraulic pistons which act on the brake pad. Of course there is friction between the pad and the piston too which helps to hold the pad in its position, but this force is neglected in this approach.

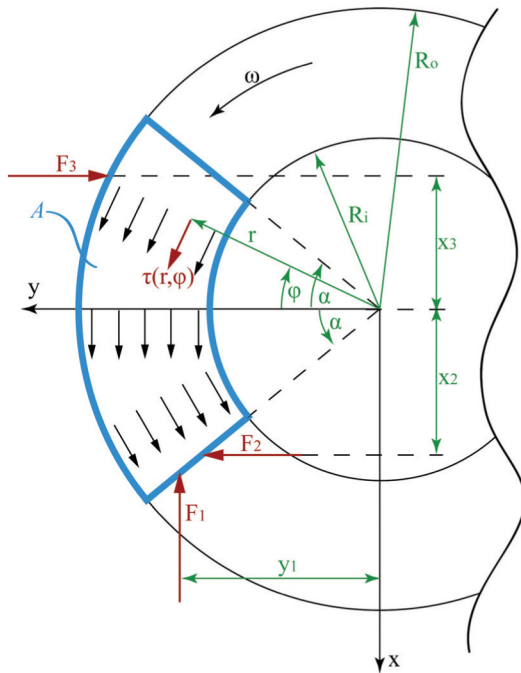


Figure 2. Principal depiction of brake disc and pad containing generic geometrical information.

Using the principles of conservation of momentum and angular momentum you can form [Equation \(1\)](#), [\(2\)](#), [\(3\)](#). In the first approximation the movement of the pad, hence the acceleration and the angular acceleration have been neglected, thus are equal to zero.

$$\int_A \tau(r, \varphi) \cos(\varphi) dA - F_1 = 0 \quad (1)$$

$$\int_A \tau(r, \varphi) \sin(\varphi) dA + F_2 - F_3 = 0 \quad (2)$$

$$-\int_A \tau(r, \varphi) r dA + F_1 y_1 + F_2 x_2 + F_3 x_3 = 0 \quad (3)$$

Furthermore it has been assumed that τ is constant in the whole contact area A , hence $\tau(r, \varphi)$ is equal to τ . Using the in [Fig. 2](#) depicted generic geometrical information about this contact area the [Equations \(1\)](#), [\(2\)](#), [\(3\)](#) can be solved for F_1 , F_2 and F_3 .

$$F_1 = 2 \tau \alpha (R_0^2 - R_i^2) \sin(\alpha) \quad (4)$$

$$F_2 = F_3 = \frac{\tau (2 \alpha R_0^3 - 3 R_0^2 y_1 \sin(\alpha) - 2 \alpha R_i^3 - 3 R_i^2 y_1 \sin(\alpha))}{3 (x_2 + x_3)} \quad (5)$$

The solved equations so far are still related to a generic brake system. In the investigated brake system the forces F_1 and F_2 load one guide pin, see [Fig. 3](#) for a principal depiction and [Fig. 4](#) for a CAD illustration. The resulting torque is supported by the locking pin (F_3).

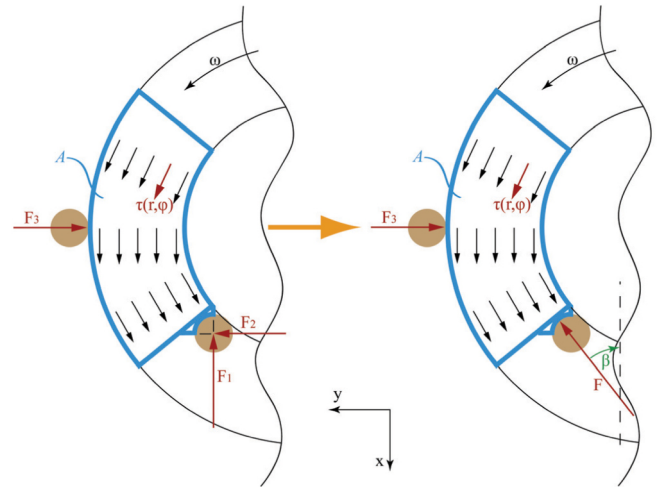


Figure 3. Analysed brake system.

[Fig. 3](#) shows the replacement of the forces F_1 and F_2 by one resulting force F and the angle β , which is defined by [Equation \(6\)](#). [Equation \(4\)](#) and [\(5\)](#) plugged in [Equation \(6\)](#) gives:

$$\beta = \arctan\left(\frac{F_2}{F_1}\right) \quad (6)$$

$$\beta = \arctan\left(\frac{2 \alpha (R_i^2 + R_i R_o + R_o^2) - 3 \gamma_1 (R_i + R_o) \sin(\alpha)}{3(R_i + R_o) (x_2 + x_3) \sin(\alpha)}\right) \quad (7)$$

At the used brake system β is equal to 8.5 angular degrees. Based on this information the two guide pins of the caliper were replaced by modified ones, on at each side of the disc. After calibration it is possible to measure the acting force F , which is depicted in [Fig. 3](#). In the area of the strain gauges the cross-section of the pin is shaped like those of an I-beam. Such modifications are necessary to have an optimal cross section and thereby an optimal signal for measuring F in the direction of β because the I-shape has a high bending section modulus and a low cross section area [\[11\]](#).

[Fig. 4](#) shows the modified guide pin, which is highlighted in red colour. The red guide pin and the blue locking pin hold the pad in position. The modifications of the red guide pin are the shape of the pin and the mounted strain gauges. Thus the acting force on the pin during braking can be measured.

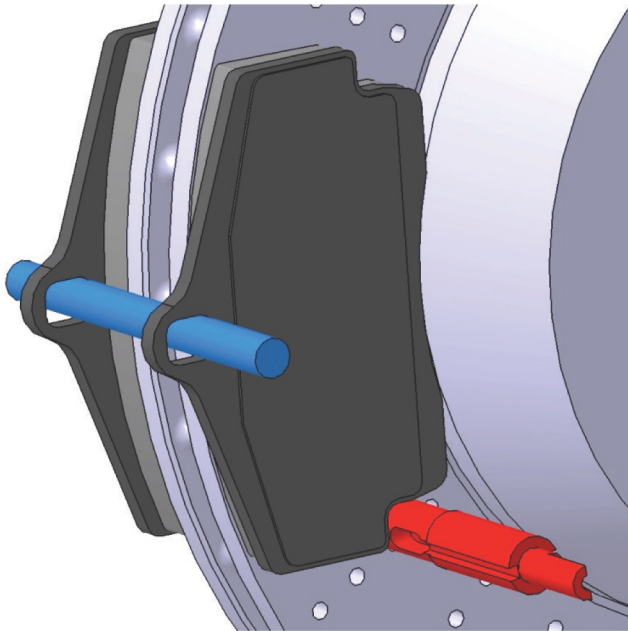


Figure 4. CAD picture of the brake system including guide and locking pin.

The red guide pin is loaded by the previously discussed force F and the blue locking pin is loaded by the force F_3 . [Fig. 5](#) depicts a photograph of the modified guide pin including the two mounted strain gauges. Two additional strain gauges are necessary for the full bridge circuit. These are mounted at the back side of the pin. Because of thermal radiation and heat conduction all four strain gauges have a thermal drift. But by

using a full bridge circuit the thermal drift can be reduced to a minimum, assuming that all strain gauges have approximately the same temperature [\[11\]](#).

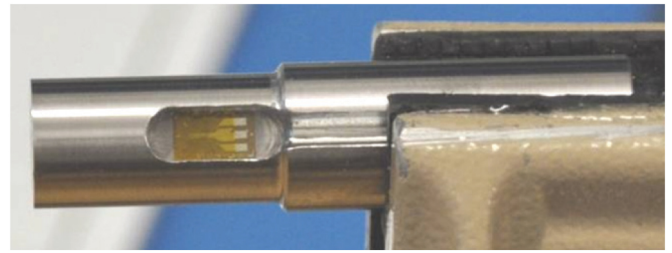


Figure 5. Modified guide pin including strain gauges.

[Fig. 6](#) shows the caliper with the two modified guide pins. Each of them is equipped with four strain gauges on it for measuring the resulting force F at the outer and the inner brake pad. Thereby the outer brake pad means the pad near the wheel and the inner one is near the knuckle of the suspension. According to SAE J2521 a microphone was mounted near the brake caliper.

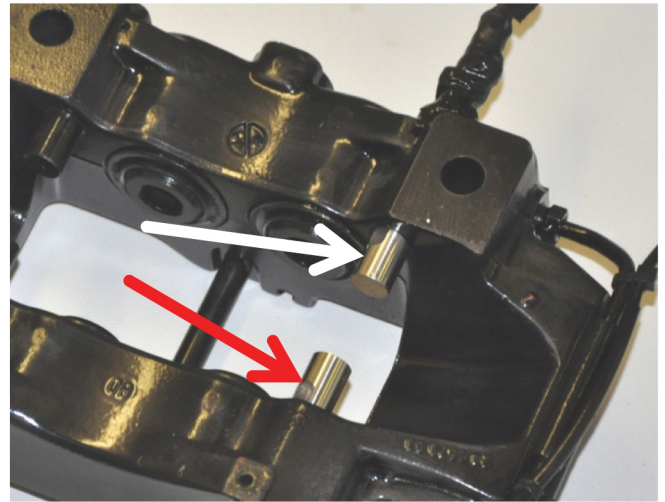


Figure 6. Brake caliper with modified guide pins, marked with arrows.

RESULTS

The main target of this research is to evaluate the potential of the new measuring system with respect to brake squeal. To simplify our analyses also a special brake pad was produced, which excludes all known countermeasures. Hence there are no damping materials or shims on the pad backplate and the contact surface has no chamfers. A brake pad manufacturer produces those prototype pads which additionally consist of sintered material. Thereby two different kinds of known poor material regarding squeal have been used. Thus it was possible to set up a brake system which squeals approximately at 70 per cent of all brake manoeuvres from

the SAE J2521 tests. Hence this system is practical for testing brake squeal measuring systems.

Fig. 7 presents an example of a brake squeal event. The test was performed with a constant rotation speed at 88 rpm and a constant brake pressure of 12 bars. The upper graph shows the data of the microphone and the lower graph the data of the two modified guide pins which are located at the inner and outer pad. Brake pressure was applied from 15.75 seconds to 19.75 seconds. The squeal event occurs from 16 to 19.75 seconds. Hence the squeal noise started right after the brake pressure was applied and stopped with the end of the brake event.

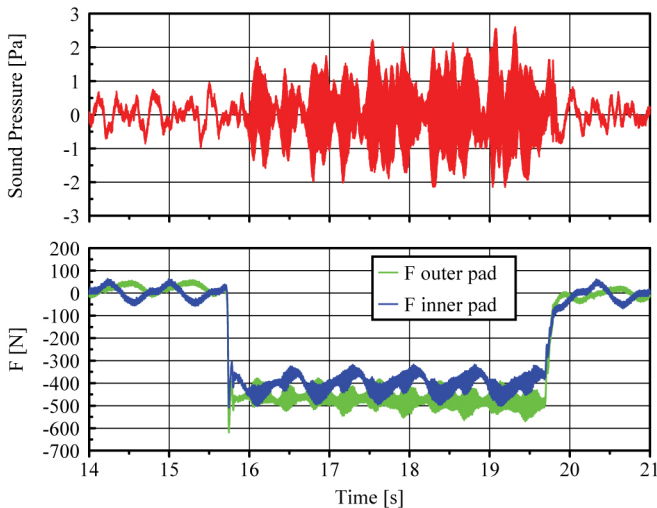


Figure 7. Measurement results for a brake squeal event.

The graphs in Fig. 8 show a zoomed section during the squeal event. In the lower graph in Fig. 8 the superposed vibrations of the forces are shown. The new developed measuring system allows measuring the frequency and the amplitude of these vibrations. The force amplitude of the outer pad is approximately 110 N at a static value of about 450 N. So for this brake squeal event the amplitude of the superposed vibration reaches up to 25 per cent of the static value. Another interesting result is that the two pads do not vibrate in phase; hence the vibrations are mirror-inverted.

Our theory is that the analysed squeal event is based on an out-of-plane vibration of the disc, which is not a “pumping” one. Of course, this system cannot determine exactly the operating deflection shape of the disc, hence the ratio of in-plane and out-of-plane vibration or the exact number of nodal diameters and nodal circles. But it probably shows which vibration type is the dominant one for the noise generation. Using a simplified finite element model of the disc the corresponding mode shapes can perhaps be defined. Conceivably it is possible to obtain information about the problematic vibration without using cost intensive measuring

systems. This conclusion will be examined with the extension of the measuring system which is explained in the outlook.

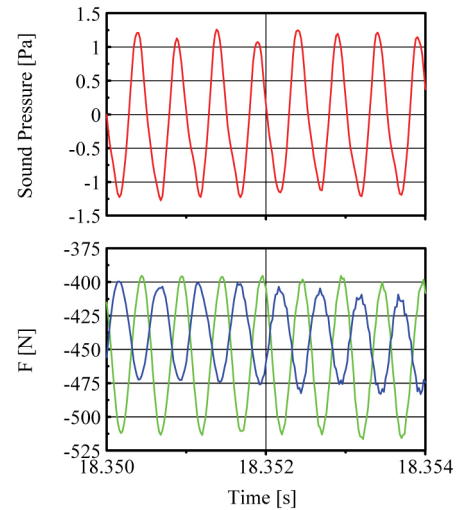


Figure 8. Zoomed signals from Fig. 7.

For further interpretations a Fast Fourier Transformation (FFT) was carried out with the three signals of Fig. 7. Therefore a *Hann-Window* was used. As Fig. 9 shows all the three signals have a significant peak at 1986 Hz. Based on the coherence of the FFTs of these three signals it can be assumed, that the measured friction force is a triggering mechanism for the noise, hence for the vibration of the disc.

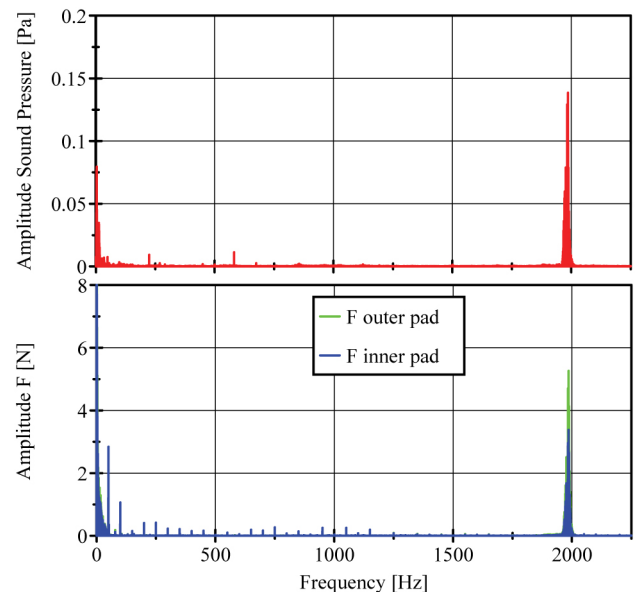


Figure 9. Fast Fourier Transform (FFT) of the three signals of Fig. 7.

In the lower graph of Fig. 7 you can also see a very low frequency vibration. This vibration depends only on the rotation speed and is caused by different reasons. One reason

is the DTV, the disc thickness variation. It can be observed, that the amplitude of the vibration and the phase shift between the two pads change slightly after e.g. high performance tests. Another reason is the dynamic unbalance in the drive shaft of the test rig. This could never be eliminated absolutely. These rotation speed depending vibrations are also visible in the FFTs in Fig. 9. A vibration induced by a rotation speed of 88 rpm is equal to 1.5 Hz.

DISCUSSION

There are lots of different friction laws discussed for friction induced vibrations. Ostermeyer [12] and Cantoni et al. [13] provide some good overviews about simple mechanical models that exhibit excitation or self excitation induced by friction. With reference to Coulomb [14] the friction force is independent of velocity. But experiments on the test rig show the typically noticed behavior, see Fig. 10 for an example of one braking maneuver. For this experiment sintered material brake pads are used, which explain the high friction coefficient.

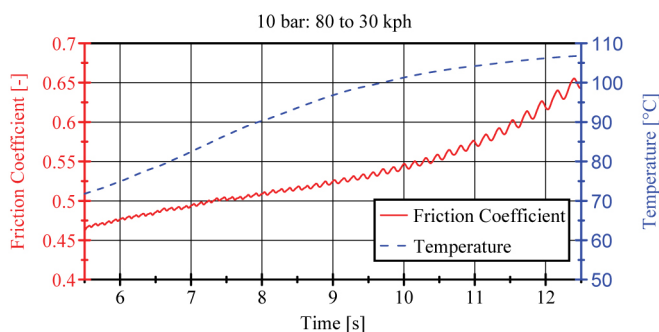


Figure 10. Example calculated friction coefficient according to SAE J2521.

As Fig. 10 shows, the friction coefficient calculated from the measured data is not constant. This is because the friction coefficient depends on several parameters, e.g. temperature, velocity, humidity, preload, wear and so on. Most models in the literature describe these different phenomena with a velocity depending friction coefficient, see e.g. [15] and [16]. Thereby mainly the negative gradient of the friction coefficient $d\mu/dv < 0$ is responsible for instability. Of course, there are also researches which use a constant friction coefficient. There the effects stick-slip and mode coupling exhibit instabilities see e.g. [17] and [18]. Fig. 11 provides you an illustration of three previous mentioned laws of friction.

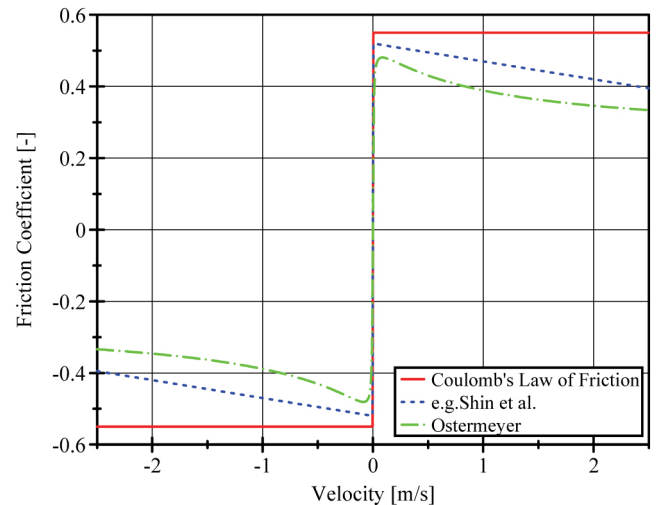


Figure 11. Example of three different laws of friction, schematic diagram adapted from [14, 16 & 19].

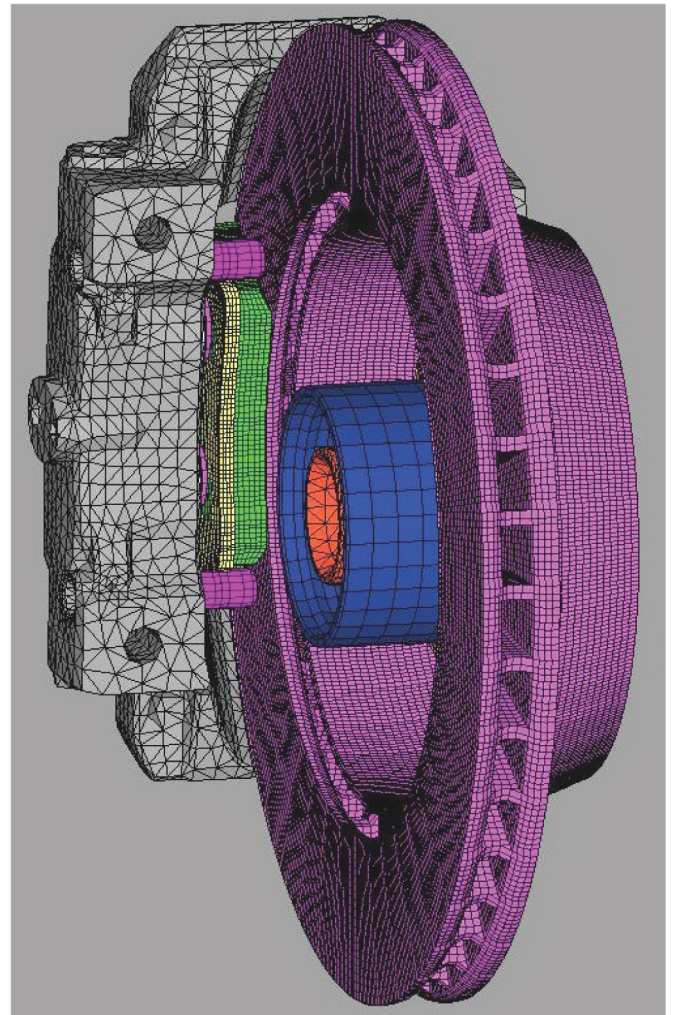


Figure 12. Finite element model of analysed axle from [20].

Fact is that the contact between disc and pad is responsible for the squeal problematic. But this contact is highly complex. There are still too many parameters which influence the exciting mechanism. However with the modified guide pins it is possible to get further insights into this tribological issue. The results of these measurements can be used to verify the laws of friction used in simulations or FEM calculations.

The existing axle has not only been analysed on road test and test rig, but also with FEM calculation. Fig. 12 shows a part of the actual finite element model. A detailed explanation and discussion have been performed in a previous work and can be found in [20].

SUMMARY

A new possibility for a measurement system has been developed. Therefore the two guide pins of the brake caliper, which hold the brake pads in position, are modified using strain gauges. These guide pins have been redesigned for reason of better signal generation. Additionally to the measurement of the friction force the sound pressure was measured with a microphone. When a squeal occurred during a brake event the main frequencies of the friction forces and the sound pressure were the same but the two friction forces did not vibrate in phase. Hence the theory was set up that the squeal is based on an out-of plane vibration of the disc.

The aim was to get a new possibility for measuring the resulting friction force and the superposed vibration. Thus it should be possible to get a further insight into the brake squeal excitation mechanism. At the moment a measuring system extension is applied to verify the new possibilities.

Due to validation reasons brake pads which consist of sintered material and without any brake squeal countermeasures have been produced. Thus it is possible to make repeatable experiments. After validation further investigations will be done in order to evaluate a contact algorithm which can be used for finite element methods calculations or analytical minimal model simulations.

OUTLOOK

The next step is an extension of the existing measurement system. Among others three miniature triaxial acceleration sensors will be mounted in the disc vents. Thus it is possible to measure the operating deflection shapes of the disc. These results will be compared with the results of the modified guide pins. The extension will also include different types of displacement and acceleration sensors, which are used to measure the vibration of the disc, the pad, the caliper and the remaining parts of the axle. This outcome will be used to verify a friction model, which will be adapted in the finite element calculation and the analytical model simulation.

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ACKNOWLEDGMENTS

The authors would like to thank Miba Friction Holding GmbH for the production of the prototype brake pads and the knowledge exchange.

The Engineering Meetings Board has approved this paper for publication. It has successfully completed SAE's peer review process under the supervision of the session organizer. This process requires a minimum of three (3) reviews by industry experts.

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